12 CHEMICAL COORDINATION IN PLANTS

Plants can detect changes in their environment and respond to them, but the responses are much slower than those in animals. This is because movements in a plant are brought about by changes in the plant's growth. This chapter is about these growth responses, and the chemicals that coordinate them.

LEARNING OBJECTIVES

- Understand that plants respond to stimuli
- Describe the phototropic response of roots and stems
- Describe the geotropic responses of roots and stems
- Understand the role of auxin in the phototropic response of stems

Chapter 6 explains how animals detect and respond to changes in their environment. Animals usually respond very quickly – for example the reflex action resulting from a painful stimulus (page 90) is over in a fraction of a second.

As in animals, some species of plant can respond rapidly to a stimulus, for example the Venus flytrap (Figure 12.1). This plant has modified leaves, which close quickly around their 'prey', trapping it. The plant then secretes enzymes to digest the insect. The movement is brought about by rapid changes in turgor of specialised cells at the base of the leaves.



▲ Figure 12.1 The Venus flytrap catches and digests insects to gain extra nutrients. The plant responds very quickly to a fly landing on one of its leaves.



▲ Figure 12.2 This bean has responded to being placed horizontally. The growing shoot has started to bend upwards.

TROPISMS

Most plants do not respond to stimuli as quickly as the Venus flytrap, because their response normally involves a change in their rate of growth. Different parts of plants may grow at different rates, and a plant may respond to a stimulus by increasing growth near the tip of its shoot or roots.

Imagine a plant growing normally in a pot. Usually, most light will be falling on the plant from above. If you turn the plant on its side and leave it for a day or so, you will see that its shoot starts to grow upwards (Figure 12.2).



▲ Figure 12.3 The shoots of these cress seedlings are showing a positive phototropism.

KEY POINT

Phototropisms are growth responses to light from one direction. Geotropisms are growth responses to the direction of gravity ('geo' refers to the Earth).

KEY POINT

Light from one direction is called 'unidirectional'. If light shines on the plant evenly from all directions, this is called 'uniform' light.

There are two stimuli acting on the plant in Figure 12.2. One is the direction of the light that falls on the plant. The other stimulus is gravity. Both light and gravity are *directional* stimuli (they act in a particular direction). The growth response of a plant to a directional stimulus is called a **tropism**. If the growth response is *towards* the direction of the stimulus, it is a *positive* tropism, and if it is *away* from the direction of the stimulus, it is a *negative* tropism. The stem of the plant in Figure 12.2 is showing a positive **phototropism** and a negative **geotropism**, which both make the stem grow upwards.

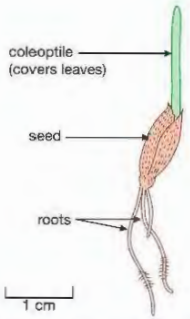
The aerial part of a plant (the 'shoot') needs light to carry out photosynthesis. This means that in most species, a positive phototropism is the strongest tropic response of the shoot. If a shoot grows towards the light, it ensures that the leaves, held out at an angle to the stem, will receive the maximum amount of sunlight. This response is easily seen in any plant placed near a window, or another source of 'one-way' or *unidirectional* light (Figure 12.3).

In darkness or uniform light, the shoot shows a negative geotropism. As you might expect, the roots of plants are strongly positively geotropic. This response makes sure that the roots grow down into the soil, where they can reach water and mineral ions. Roots also fix a plant firmly in place – they provide 'anchorage'.

The roots of some species that have been studied are also negatively phototropic, but most roots don't respond to directional light at all. Table 12.1 summarises the main tropisms.

Table 12.1: The main responses of plants to directional stimuli (tropisms).

Stimulus	Name of response	Response of shoots	Response of roots
light	phototropism	grow towards light source (positive phototropism)	most species show no response; some grow away from light (negative phototropism)
gravity	geotropism	grow away from direction of gravity (negative geotropism)	grow towards direction of gravity (positive geotropism)



▲ Figure 12.4 Germinating oat seedling

EXTENSION WORK

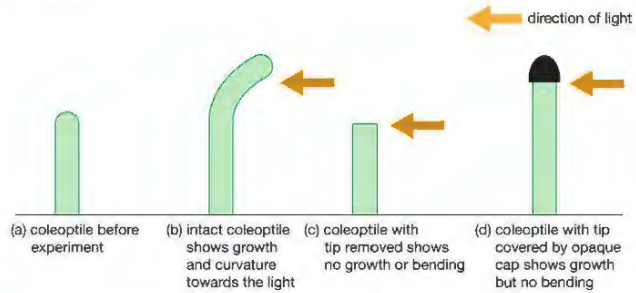
Charles Darwin is best known for his theory of evolution by natural selection (Chapter 19). However, he carried out research in other areas of biology, including studying plant hormones. You could do some research to find out more about his life and work. Make a list of his achievements.

DETECTING THE LIGHT STIMULUS – PLANT HORMONES

Plants do not have the obvious sense organs and nervous system of animals. However, since they respond to stimuli such as light and gravity, they must have some way of detecting them and coordinating the response. The detection system of phototropism was first investigated by the great English biologist Charles Darwin (see Chapter 19) in the late nineteenth century. Instead of using stems, Darwin (and later scientists) used cereal coleoptiles, which are easier to grow and use in experiments.

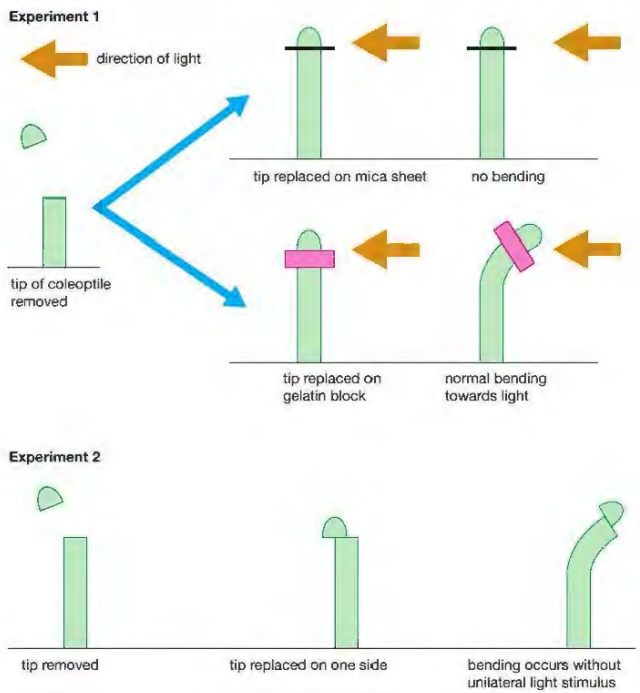
A **coleoptile** is a protective sheath that covers the first leaves of a cereal seedling. It protects the delicate leaves as the shoot emerges through the soil (Figure 12.4). Coleoptiles have a simple structure and are easy to grow, so they are often used to investigate tropisms.

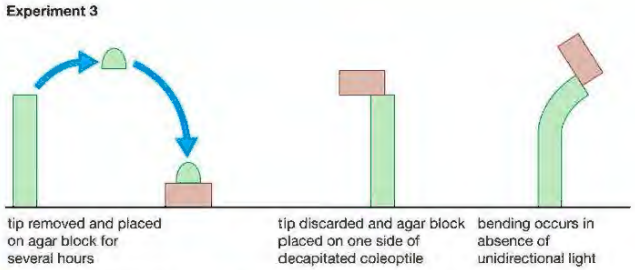
Darwin showed that the stimulus of unidirectional light was detected by the tip of the coleoptile, and transmitted to a growth zone, just behind the tip (Figure 12.5).



▲ Figure 12.5 Darwin's experiments with phototropism (1880)

Since plants don't have a nervous system, biologists began to look for a chemical messenger (or plant **hormone**) that might be the cause of phototropism in coleoptiles. Between 1910 and 1926 several scientists investigated this problem. Some of their results are summarised in Figure 12.6.





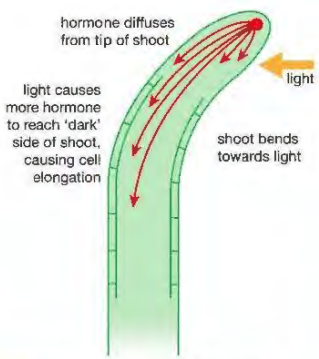
▲ Figure 12.6 Experiments on coleoptiles that helped to explain the mechanism of phototropism.

In experiment 1, the stimulus for growth was found to pass through materials such as gelatine, which absorbs water-soluble chemicals, but not through materials such as mica (a mineral), which is impermeable to water. This made biologists think that the stimulus was a chemical that was soluble in water.

In experiment 2, it was shown that the phototropic response could be brought about, even *without* unidirectional light, by removing a coleoptile tip ('decapitating' the coleoptile) and placing the tip on one side of the decapitated stalk.

In experiment 3, it was found that the hormone could be collected in another water-absorbing material (a block of agar jelly). Placing the agar block on one side of the decapitated coleoptile stalk caused it to bend.

Investigations like experiments 2 and 3 led scientists to believe that the hormone caused bending by stimulating growth on the side of the coleoptile furthest from the light. The theory is that the hormone is produced in the tip of the shoot, and diffuses back down the shoot. If the shoot is in the dark, or if light is all around the shoot, the hormone diffuses at equal rates on each side of the shoot, so it stimulates the shoot equally on all sides. However, if the shoot is receiving light from one direction, the hormone moves away from the light as it diffuses downwards. The higher concentration of hormone on the 'dark' side of the shoot stimulates cells there to grow, making the shoot bend towards the light (Figure 12.7).



▲ Figure 12.7 How movement of a plant hormone causes phototropism.

Since these experiments were carried out, scientists have identified the hormone responsible. It is called **auxin**. Several other types of plant hormone have been found. Like auxin, they all influence growth and development of plants in one way or another, so that many scientists prefer to call them **plant growth substances** rather than plant hormones.

EXTENSION WORK

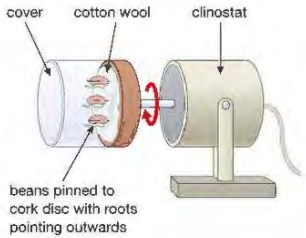
'Auxin' should really be 'auxins', since there are a number of chemicals with very similar structures making up a group of closely related plant hormones.

EXTENSION WORK

As well as auxins, there are four other main groups of plant hormones, called gibberellins, cytokinins, abscisic acid and ethene. They control many aspects of plant growth and development, apart from tropisms. These include growth of buds, leaves and fruit, fruit ripening, seed germination, leaf fall and opening of stomata, to name just a few.

USING A CLINOSTAT TO SHOW TROPISMS

A clinostat is a piece of apparatus consisting of an electric motor turning a cork disc. Germinating seeds are attached to the disc. The motor turns the disc and seeds around very slowly, so that the movement eliminates the effect of any directional stimulus that may be acting on the seeds. The clinostat can be turned through 90°, so that the disc rotates either horizontally or vertically.



▲ Figure 12.8 A clinostat

The clinostat can be used to demonstrate tropisms, for example geotropism in roots. A few beans are soaked in water overnight and then placed on wet cotton wool for a day or two, until the first root of each seed (called the radicle) grows to a length of about 2 cm.

Wet cotton wool is attached to the cork disc of two clinostats. Three or four of the germinating bean seeds are pinned on to each of the discs, with their radicles pointing outwards. Covers are placed over them to keep the air around the beans moist.

The clinostats are turned on their sides, as shown in Figure 12.8. One clinostat is switched on, with the other left switched off, as a control. Both clinostats are left set up for a few days. The cotton wool is watered regularly to keep it damp.

After this time, the radicles of the beans on the control clinostat will have grown downwards, under the influence of gravity. Those on the moving clinostat will have grown straight out horizontally. The continuously changing direction of the gravitational stimulus acting on the seeds of the moving clinostat cancels out the geotropic response.

CHAPTER QUESTIONS

More questions on chemical coordination in plants can be found at the end of Unit 3 on page 182.

SKILLS CRITICAL THINKING (5)

1 Which of the following statements about tropisms is correct?

A Roots show positive phototropism and negative geotropism
B Roots show negative phototropism and positive geotropism
C Stems show negative phototropism and negative geotropism
D Stems show positive phototropism and positive geotropism

(6)

2 Below are three statements about phototropism in a coleoptile.

1. Auxin is made in the tip of the coleoptile
2. Auxin diffuses towards the light
3. A high concentration of auxin stimulates growth of the coleoptile towards the light.

Which of these statements are true?

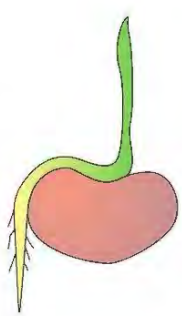
A 1 and 2
B 1 and 3
C 2 and 3
D 1, 2 and 3

(5)

3 The diagram below shows a bean seedling that has been growing in the dark.

What is the name given to the response of the shoot?

A positive phototropism
B negative phototropism
C positive geotropism
D negative geotropism



SKILLS CRITICAL THINKING (5)

4 Movement of auxin is thought to take place by which of the following methods?

A diffusion
B active transport
C osmosis
D mass flow in the phloem

5 a What are the main stimuli affecting the growth of
i the shoot? ii the root?

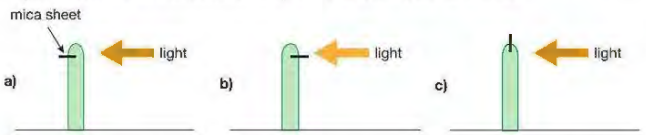
(6) b How does a plant benefit from a positive phototropism in its stem?

SKILLS INTERPRETATION (7)

6 Draw a labelled diagram to show how auxin brings about phototropism in a coleoptile that has light shining on it from one direction.

SKILLS REASONING (8)

7 An experiment was carried out to investigate phototropism in a coleoptile. The diagram shows what was done. Predict the results you would expect to get in each of the experiments a) to c). Explain your answers.

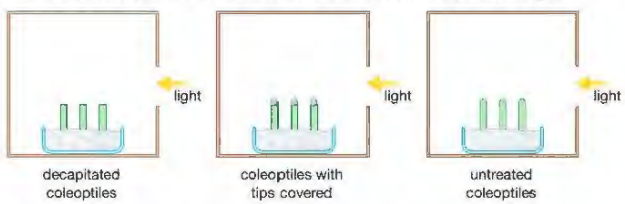


SKILLS EXECUTIVE FUNCTION (8)

8 An experiment was carried out to investigate which part of a coleoptile is sensitive to a light stimulus. Three dishes containing wheat seedlings were subjected to different treatments:

- Dish 1 – 2 mm was removed from the growing coleoptiles ('decapitated' coleoptiles)
- Dish 2 – the tips of the coleoptiles were covered with caps of aluminium foil
- Dish 3 – the seedlings were left untreated as a control.

The coleoptiles were measured and the mean length of each treatment group was calculated. Each batch of seedlings was then placed in a box with a hole cut in one side to provide a unidirectional source of light.



After 48 hours the new mean length of the coleoptiles was found for each treatment group, and the percentage increase in length calculated. Any change in direction of growth of the coleoptiles was noted.

SKILLS REASONING

a Suggest which treatment would produce the greatest percentage increase in length and which the least. Explain your answer.

b What differences in direction of growth would you expect to see as a result of the three treatments? Explain your answer.

(5) c Explain why it is necessary to calculate the *percentage* increase in the length of the coleoptiles, rather than simply the increase in length.